6b. Named Tuples and Dictionaries

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INTRODUCTION

Our previous discussions of collections have centered around vectors. We also introduced tuples, although without exploring them in much depth. The current section fills that gap y offering a more comprehensive analysis of tuples.

We then broaden the picture by introducing two additional collection types: **named tuples** and **dictionaries**. Along the way, we'll develop a general perspective on collections in terms of their keys and values. Additionally, we'll discuss common operations for manipulating collections, and examine techniques for transforming one collection into another.

KEYS AND VALUES

Most collections in Julia are characterized by **keys**. They serve as unique identifiers for their elements and are paired with a corresponding **value**. [1] For instance, the vector `x = [2, 4, 6]` has the indices `1, 2, 3` as its keys, and `2, 4, 6` as their respective values.

Keys are more general than indices: while indices are limited to integer identifiers, keys can be any valid Julia object (e.g., strings, numbers, or other objects).

To extract the keys and values of a collection, Julia provides the functions `keys` and `values`. The following examples demonstrate their behavior with vectors and tuples, whose keys are represented by indices. Note that neither `keys` nor `values` return a vector, requiring the `collect` function to obtain a vector representation.

VECTOR

```
x        = [4, 5, 6]

x_keys   = collect(keys(x))
x_values = collect(values(x))
```

```
julia> x_keys
3-element Vector{Int64}:
 1
 2
 3
julia> x_values
3-element Vector{Int64}:
 4
 5
 6
```

TUPLE

```
x        = (4, 5, 6)

x_keys   = collect(keys(x))
x_values = collect(values(x))
```

```
julia> x_keys
3-element Vector{Int64}:
 1
 2
 3
julia> x_values
3-element Vector{Int64}:
 4
 5
 6
```

THE TYPE `PAIR`

Collections of key-value pairs in Julia are represented by the type `Pair{<key type>, <value type>}`. Although we'll rarely work with objects of this type in this book, they form the basis for constructing other collections, such as dictionaries and named tuples.

A key-value pair can be created by using the operator `=>`, as in `<key> => <value>`. For instance, `"a" => 1` constructs a pair, where `a` is the key and `1` the corresponding value. Alternatively, pairs can be created using the function `Pair(<key>, <value>)`, where `Pair("a",1)` is equivalent to `"a" => 1`.

Given a pair `x`, we can access its key via either `x[1]` or `x.first`. Likewise, its value can be obtained using either `x[2]` or `x.second`. All this is demonstrated below.

KEY OF A PAIR

```
some_pair = ("a" => 1)       # or simply 'some_pair = "a" => 1'

some_pair = Pair("a", 1)     # equivalent
```

```
julia> some_pair
"a" => 1
julia> some_pair[1]
"a"
julia> some_pair.first
"a"
```

VALUE OF A PAIR

```
some_pair = ("a" => 1)       # or simply 'some_pair = "a" => 1'

some_pair = Pair("a", 1)     # equivalent
```

```
julia> some_pair
"a" => 1
julia> some_pair[2]
1
julia> some_pair.second
1
```

THE TYPE `SYMBOL`

The type used to represent keys vary across collections. One particularly common choice is `Symbol`, which offers an efficient way to encode string-based identifiers. A symbol named `x` is written as `:x`, and can be constructed from a string using the function `Symbol(<string>)`.[2]

A VECTOR OF SYMBOLS

```
vector_symbols = [:x, :y]
```

```
julia> vector_symbols
2-element Vector{Symbol}:
 :x
 :y
```

A VECTOR OF SYMBOLS - EQUIVALENT

```
vector_symbols = [Symbol("x"), Symbol("y")]
```

```
julia> vector_symbols
2-element Vector{Symbol}:
 :x
 :y
```

NAMED TUPLES

> ⚠️ **Warning!**
>
> As we'll show **later in the book**, **tuples and named tuples are only suitable for small collections**. Using them with large collections can result in poor performance or even fatal errors. For large collections, arrays remain the preferred choice.
>
> Defining what qualifies as *small* is challenging, and unfortunately there's no definitive answer. We can only indicate that collections with fewer than 10 elements are certainly small, while those exceeding 100 elements clearly exceed the intended use.

Named tuples share several properties with regular tuples, including their **immutability**. However, they also exhibit some notable differences. An important one is that **the keys of named tuples are objects of type** `Symbol`, in contrast to the numerical indices used for regular tuples.

Named tuples also differ syntactically, requiring being enclosed in parentheses `()`. Omitting them isn't possible, unlike with regular tuples. Furthermore, when creating a single-element named tuple, the syntax requires either a trailing comma `,` after the element (similar to regular tuples) or a leading semicolon `;` before the element.[3]

To construct a named tuple, each element must be specified in the format `<key> = <value>`, such as `a = 10`. Alternatively, we can use a pair `<key with Symbol type> => <value>`, as in `:a => 10`. Once a named tuple `nt` is created, the element `a` can be accessed either by key lookup `nt[:a]` or by dot syntax `nt.a`.

The following examples illustrate these concepts.

NAMED TUPLE

```julia
# all 'nt' are equivalent
nt = (  a=10, b=20)
nt = (; a=10, b=20)
nt = (  :a => 10, :b => 20)
nt = (; :a => 10, :b => 20)
```

```julia
julia> nt
(a = 10, b = 20)

julia> nt.a
10

julia> nt[:a]
10
```

SINGLE-ELEMENT NAMED TUPLE

```
# all 'nt' are equivalent
nt = (  a=10,)
nt = (; a=10 )
nt = (  :a => 10,)
nt = (; :a => 10 )

#not 'nt =  (a = 10)'  -> this is interpreted as 'nt = a = 10'
#not 'nt = (:a => 10)' -> this is interpreted as a pair
```

```
julia> nt
(a = 10,)
julia> nt.a
10
julia> nt[:a]
10
```

Remark

To see the list of keys and values, we can employ the functions `keys` and `values`.

KEYS AND VALUES OF NAMED TUPLE

```
nt        = (a=10, b=20)

nt_keys   = collect(keys(nt))
nt_values = collect(values(nt))
```

```
julia> nt_keys
2-element Vector{Symbol}:
 :a
 :b
julia> nt_values
2-element Vector{Int64}:
 10
 20
```

DISTINCTION BETWEEN THE CREATION OF TUPLES AND NAMED TUPLES

It's possible to create named tuples from existing variables. For instance, given variables `x = 10` and `y = 20`, one can define `nt = (; x, y)`. This creates a named tuple with keys `x` and `y`, whose corresponding values are `10` and `20`.

The semicolon `;` plays a crucial role in this construction, as it distinguishes named tuples from regular tuples. Omitting it, as in `nt = (x, y)`, would instead result in a regular tuple.

MULTIPLE ELEMENTS

```
x   = 10
y   = 20

nt  = (; x, y)
tup = (x, y)
```

```
julia> nt
(x = 10, y = 20)
julia> tup
(10, 20)
```

SINGLE ELEMENT

```
x   = 10


nt  = (; x)
tup = (x, )
```

```
julia> nt
(x = 10,)
julia> tup
(10,)
```

DICTIONARIES

Dictionaries are collections of key-value pairs, exhibiting three distinctive features:

- **Dictionary keys can be any object**: strings, numbers, and other objects are possible.
- **Dictionaries are mutable**: elements can be modified, added, and removed after creation.
- **Dictionaries are unordered**: keys have no inherent order.

The function `Dict` can be used to create dictionaries, where each argument is a key-value pair written in the form `<key> => <value>`.

NUMBER

```
some_dict = Dict(3 => 10, 4 => 20)
```

```
julia> some_dict
Dict{Int64, Int64} with 2 entries:
  4 => 20
  3 => 10
julia> some_dict[1]
10
```

STRING

```
some_dict = Dict("a" => 10, "b" => 20)
```

```
julia> some_dict
Dict{String, Int64} with 2 entries:
  "b" => 20
  "a" => 10
julia> some_dict["a"]
10
```

SYMBOL

```
some_dict = Dict(:a => 10, :b => 20)
```

```
julia> some_dict
Dict{Symbol, Int64} with 2 entries:
  :a => 10
  :b => 20
julia> some_dict[:a]
10
```

TUPLE

```
some_dict = Dict((1,1) => 10, (1,2) => 20)
```

```
julia> some_dict
10
julia> some_dict[(1,1)]
10
```

Regular dictionaries are inherently unordered, meaning that the access to their elements doesn't follow any pattern. The following example illustrates this, by collecting the dictionary keys into a vector.[4]

NUMBER

```
some_dict      = Dict(3 => 10, 4 => 20)

keys_from_dict = collect(keys(some_dict))
```

```
julia> keys_from_dict
2-element Vector{Int64}:
 4
 3
```

STRING

```
some_dict      = Dict("a" => 10, "b" => 20)

keys_from_dict = collect(keys(some_dict))
```

```
julia> keys_from_dict
2-element Vector{String}:
 "b"
 "a"
```

SYMBOL

```
some_dict      = Dict(:a => 10, :b => 20)

keys_from_dict = collect(keys(some_dict))
```

```
julia> keys_from_dict
2-element Vector{Symbol}:
 :a
 :b
```

TUPLE

```
some_dict      = Dict((1,1) => 10, (1,2) => 20)

keys_from_dict = collect(keys(some_dict))
```

```
julia> keys_from_dict
2-element Vector{Tuple{Int64, Int64}}:
 (1, 2)
 (1, 1)
```

CREATING TUPLES, NAMED TUPLES, AND DICTIONARIES

Tuples, named tuples, and dictionaries can be constructed from other collections. The only requirement is that the source collection possesses a key-value structure.

To demonstrate this feature, we begin by creating **dictionaries** from a variety of collections.

VECTOR

```
vector = [10, 20] # or tupl = (10,20)



dict = Dict(pairs(vector))
```

```
julia> dict
Dict{Int64, Int64} with 2 entries:
  2 => 20
  1 => 10
```

VECTORS

```
keys_for_dict   = [:a, :b]
values_for_dict = [10, 20]


dict = Dict(zip(keys_for_dict, values_for_dict))
```

```
julia> dict
Dict{Symbol, Int64} with 2 entries:
  :a => 10
  :b => 20
```

TUPLES

```
keys_for_dict   = (:a, :b)
values_for_dict = (10, 20)


dict = Dict(zip(keys_for_dict, values_for_dict))
```

```
julia> dict
Dict{Symbol, Int64} with 2 entries:
  :a => 10
  :b => 20
```

NAMED TUPLE

```
nt_for_dict = (a = 10, b = 20)



dict = Dict(pairs(nt_for_dict))
```

```
julia> dict
Dict{Symbol, Int64} with 2 entries:
  :a => 10
  :b => 20
```

COMPREHENSION

```
keys_for_dict      = (:a, :b)
values_for_dict    = (10, 20)
vector_keys_values     =     [(keys_for_dict[i],     values_for_dict[i])     for     i     in
eachindex(keys_for_dict)]

dict = Dict(vector_keys_values)
```

```
julia> dict
Dict{Symbol, Int64} with 2 entries:
  :a => 10
  :b => 20
```

Likewise, we can define a **tuple** from other collections, as shown below.

VARIABLES

```
a              = 10
b              = 20

tup            = (a, b)
```

```
julia> tup
(10, 20)
```

VECTORS 1

```
values_for_tup = [10, 20]

tup            = (values_for_tup... ,)
```

```
julia> tup
(10, 20)
```

VECTORS 2

```
values_for_tup = [10, 20]

tup            = Tuple(values_for_tup)
```

```
julia> tup
(10, 20)
```

Finally, **named tuples** can also be constructed from other collections.

VARIABLES

```
a = 10
b = 20


nt = (; a, b)
```

```
julia> nt
(a = 10, b = 20)
```

VECTORS 1

```
keys_for_nt   = [:a, :b]
values_for_nt = [10, 20]


nt = (; zip(keys_for_nt, values_for_nt)...)
```

```
julia> nt
(a = 10, b = 20)
```

VECTORS 2

```
keys_for_nt   = [:a, :b]
values_for_nt = [10, 20]


nt = NamedTuple(zip(keys_for_nt, values_for_nt))
```

```
julia> nt
(a = 10, b = 20)
```

TUPLES

```
keys_for_nt   = (:a, :b)
values_for_nt = (10, 20)


nt = NamedTuple(zip(keys_for_nt, values_for_nt))
```

```
julia> nt
(a = 10, b = 20)
```

COMPREHENSION

```
keys_for_nt        = [:a, :b]
values_for_nt      = [10, 20]
vector_keys_values = [(keys_for_nt[i], values_for_nt[i]) for i in eachindex(keys_for_nt)]

nt = NamedTuple(vector_keys_values)
```

```
julia> nt
(a = 10, b = 20)
```

DICTIONARY

```
dict = Dict(:a => 10, :b => 20)



nt = NamedTuple(vector_keys_values)
```

```
julia> nt
(a = 10, b = 20)
```

DESTRUCTURING TUPLES AND NAMED TUPLES

Previously, we demonstrated how to create tuples and named tuples from variables. Next, we show that the reverse operation is also possible, where **values are extracted from a tuple or named tuple and assigned to separate variables**. This process is known as **destructuring**, enabling users to "unpack" the values of a collection into distinct variables.

Destructuring involves the assignment operator `=` with either a tuple or named tuple on the left-hand side. The choice between one or the other determines what objects can be used on the right-hand side. Specifically, tuples on the left-hand side are quite flexible, allowing values to be unpacked from a variety of collections. Named tuples on the left-hand side, instead, necessarily require a named tuple on the right-hand side. Next, we develop each case separately.

DESTRUCTURING COLLECTIONS THROUGH TUPLES

Given a collection `list` with two elements, destructuring via tuples allows us to unpack its values into the variables `x` and `y`. The syntax for this is `<tuple> = <collection>`, as in `x,y = list`. In the following, we illustrate the process by considering different objects as `list`.

VECTORS

```
list = [3,4]

x,y  = list
```

```
julia> x
3
julia> y
4
```

RANGES

```
list = 3:4

x,y  = list
```

```
julia> x
3
julia> y
4
```

TUPLES

```
list = (3,4)

x,y  = list
```

```
julia> x
3
julia> y
4
```

NAMED TUPLES

```
list = (a = 3, b = 4)

x,y  = list
```

```
julia> x
3
julia> y
4
```

In addition to unpacking all elements, destructuring can also be applied to only a subset of elements. The assignment is then carried out sequentially, following the collection's inherent order.

Importantly, this method doesn't allow skipping specific values. When a value must be disregarded, the conventional approach is to bind it to the special variable name `_`. This symbol serves purely as a placeholder, indicating that the value is unimportant and has no effect on execution.

For illustration, we'll use a vector as an example of `list`. Nonetheless, the same principle applies to any collection.

FIRST ELEMENT

```
list  = [3,4,5]

(x,)  = list
```

```
julia> x
3
```

FIRST ELEMENTS

```
list  = [3,4,5]

x,y   = list
```

```
julia> x
3
julia> y
4
```

LAST ELEMENT

```
list  = [3,4,5]

_,_,z = list        # _ or any symbol (_ just signals we don't care about that value)
```

```
julia> z
5
```

FIRST AND LAST ELEMENT

```
list  = [3,4,5]

x,_,z = list        # _ or any symbol (it just signals we don't care about that value)
```

```
julia> x
3
julia> y
5
```

DESTRUCTURING WITH NAMED TUPLES ON BOTH SIDES

Destructuring can also be applied with named tuples on the left-hand side of an assignment. In this form, values are extracted by referencing directly to their field names, rather than relying on their positional order. The main advantage of this approach is flexibility: variables can be assigned in any order, provided each name matches a field in the named tuple.

MULTIPLE VARIABLES

```
nt              = (; key1 = 10, key2 = 20, key3 = 30)

(; key3, key1) = nt            # keys in any order
```

```
julia> key1
10
julia> key3
30
```

A SINGLE VARIABLE

```
nt              = (; key1 = 10, key2 = 20, key3 = 30)

(; key2)        = nt            # only one key
```

```
julia> key2
20
```

Remark

When destructuring with a tuple on the left-hand side and a named tuple on the right-hand side, keep in mind that the assignment is carried out strictly by position. This means that variable names on the left don't influence the result. In other words, the keys of the named tuple are completely ignored during the process.

NAMED TUPLE

```
nt = (; key1 = 10, key2 = 20, key3 = 30)

 key2, key1      = nt        # variables defined according to
POSITION
(key2, key1)   = nt     # alternative notation
```

```
julia> key2
10
julia> key1
20
```

TUPLE

```
nt = (; key1 = 10, key2 = 20, key3 = 30)

(; key2, key1) = nt      # variables defined according to KEY
 ; key2, key1  = nt      # alternative notation
```

```
julia> key2
20
julia> key1
10
```

The same caveat applies to assignments of single variables.

NAMED TUPLE

```
nt       = (; key1 = 10, key2 = 20)

(key2,)   = nt                 # variable defined according to
POSITION
```

```
julia> key2
10
```

TUPLE

```
nt       = (; key1 = 10, key2 = 20)

(; key2) = nt            # variable defined according to KEY
```

```
julia> key2
20
```

APPLICATIONS OF DESTRUCTURING

Destructuring named tuples is particularly valuable in scientific modelling, where numerous parameters are referenced repeatedly throughout the code. By grouping all parameters into a single named tuple, we can pass them to a function as *one* consolidated argument. Functions written in this style typically begin by destructuring the named tuple, extracting exactly the parameters needed for the computations.

MULTIPLE ARGUMENTS

```julia
β = 3
δ = 4
ϵ = 5

# function 'foo' uses β and δ, but not ϵ
function foo(x, δ, β)
    x * δ + exp(β) / β


end

output = foo(1, δ, β)
```

```
julia> output
10.6952
```

NO DESTRUCTURING

```julia
parameters_list = (; β = 3, δ = 4, ϵ = 5)



# function 'foo' uses β and δ, but not ϵ
function foo(x, parameters_list)
    x * parameters_list.δ + exp(parameters_list.β) / parameters_list.β


end

output = foo(1, parameters_list)
```

```
julia> output
10.6952
```

DESTRUCTURING

```julia
parameters_list = (; β = 3, δ = 4, ϵ = 5)



# function 'foo' uses β and δ, but not ϵ
function foo(x, parameters_list)
    (; β, δ) = parameters_list

    x * δ + exp(β) / β
end

output = foo(1, parameters_list)
```

```
julia> output
10.6952
```

Destructuring also provides a convenient solution for retrieving multiple outputs from a function. When a function wraps its results in a tuple, we can then immediately unpack the tuple into separate variables. This often makes the subsequent code easier to read and reason about. In the example below, the function `foo` returns a tuple, which is then unpacked into variables `x`, `y`, and `z`.

TUPLE AS OUTPUT

```
function foo()
    out1 = 2
    out2 = 3
    out3 = 4

    out1, out2, out3
end

x, y, z = foo()
```

Another common use of destructuring arises when we only need a subset of a function's outputs. While both tuples and named tuples can be applied for this purpose, tuples offer greater flexibility, as they can be combined with various types of collections. In contrast, named tuples are restricted to returning another named tuple as the function's output, thus requiring knowing the field names in advance.

The following example illustrates this functionality by extracting the first and third output of `foo`, while ignoring the second.

TUPLE AS OUTPUT (TUPLE)

```
function foo()
    out1 = 2
    out2 = 3
    out3 = 4

    out1, out2, out3
end

x, _, z = foo()
```

NAMED TUPLE AS OUTPUT (NAMED TUPLES)

```
function foo()
    out1 = 2
    out2 = 3
    out3 = 4

    (; out1, out2, out3)
end

(; out1, out3) = foo()
```

FOOTNOTES

[1.] Not all collections map keys to values. For example, the type `Set`, which represents a group of unique unordered elements, doesn't have a key-value structure.

[2.] `Symbol` also enables the programmatic creation of variables. A typical use case arises in data analysis, where symbols are employed to generate new columns in tabular data structures.

[3.] The semicolon notation `;` may seem odd, but it actually comes from the syntax for keyword arguments in functions.

[4.] The package `OrderedCollections` addresses this, by offering a special dictionary called `OrderedDict`. It behaves similarly to regular dictionaries, including their syntax, but endows the dictionary with an order.